

The data scientist at an online store is trying to determine how many people use the website during 08:00 and 17:00 for different days of the week. She determines that weekends (Saturdays and Sundays) are busier than weekdays. In the week, people sign-on to the website at an average rate of 35 people per hour. On the weekends, people sign-on to the website at an average rate of 63 people per hour.

Let X be the number of people who sign on to the website.

Let W be that it is a weekday and W' be that it is a weekend. Thus $P(W) = \frac{5}{7}$ and $P(W') = \frac{2}{7}$

$$\lambda_{\text{weekday}} = 35 \text{ per hour and } \frac{35}{60} \text{ per minute}$$

$$\lambda_{\text{weekend}} = 63 \text{ per hour and } \frac{63}{60} \text{ per minute}$$

1. Suppose you randomly select a day of the week. What is the probability that three people sign-on to the website between 08:00 and 08:01?

$$\begin{aligned} P(X = 3) &= P(X = 3 \cap \text{weekday}) + P(X = 3 \cap \text{weekend}) \\ &= P(X = 3 | \lambda_{\text{weekday}})P(W) + P(X = 3 | \lambda_{\text{weekend}})P(W') \\ &= \left[\frac{e^{-\frac{35}{60}} \left(\frac{35}{60}\right)^3}{3!} \right] \left(\frac{5}{7}\right) + \left[\frac{e^{-\frac{63}{60}} \left(\frac{63}{60}\right)^3}{3!} \right] \left(\frac{2}{7}\right) \\ &= 0.032477 \\ &\approx 0.0325 \end{aligned}$$

2. Suppose you randomly select a day of the week. What is the probability that 37 people sign-on to the website between 16:00 and 17:00?

$$\begin{aligned} P(X = 37) &= P(X = 37 \cap \text{weekday}) + P(X = 37 \cap \text{weekend}) \\ &= P(X = 37 | \lambda_{\text{weekday}})P(W) + P(X = 37 | \lambda_{\text{weekend}})P(W') \\ &= \left[\frac{e^{-35} (35)^3}{3!} \right] \left(\frac{5}{7}\right) + \left[\frac{e^{-63} (63)^3}{3!} \right] \left(\frac{2}{7}\right) \\ &= 0.0442263 \\ &\approx 0.0442 \end{aligned}$$

3. Suppose that for some **given** day 40 people sign-on to the website during 13:00 and 14:00. What is the probability that it is a weekday (Monday - Friday)? [Hint: Consider Bayes' theorem.]

$$P(X = 40 | W) = \frac{e^{-35} 35^{40}}{40!} = 0.044748 \approx 0.0447$$

$$P(X = 40 | W') = \frac{e^{-63} 63^{40}}{40!} = 0.000503 \approx 0.0005$$

$$\begin{aligned}
P(W|X=40) &= \frac{P(X=40 \cap \text{weekday})}{P(X=40)} \\
&= \frac{P(X=40|W)P(W)}{P(X=40|W)P(W) + P(X=40|W')P(W')} \\
&= \frac{0.0447 \times \frac{5}{7}}{(0.0447 \times \frac{5}{7}) + (0.0005 \times \frac{2}{7})} \\
&= \frac{0.0320}{0.0320 \times 0.0001} \\
&= 0.995525268 \\
&\approx 0.9955
\end{aligned}$$

4. Suppose that for some **given** day 40 people sign-on to the website during 13:00 and 14:00. What is the probability that it is a weekend (Saturday or Sunday)? [Hint: Consider Bayes' theorem.]

$$P(X=40|W) = \frac{e^{-35}35^{40}}{40!} = 0.044748 \approx 0.0447$$

$$P(X=40|W') = \frac{e^{-63}63^{40}}{40!} = 0.000503 \approx 0.0005$$

$$\begin{aligned}
P(W'|X=40) &= \frac{P(X=40 \cap \text{weekend})}{P(X=40)} \\
&= \frac{P(X=40|W')P(W')}{P(X=40|W)P(W) + P(X=40|W')P(W')} \\
&= \frac{0.0005 \times \frac{2}{7}}{(0.0447 \times \frac{5}{7}) + (0.0005 \times \frac{2}{7})} \\
&= \frac{0.0001}{0.0320 \times 0.0001} \\
&= 0.00447473 \\
&\approx 0.0045
\end{aligned}$$